Air Quality, Energy and Health
Science and Policy Summaries

Air quality monitoring for enhanced understanding of health impacts and beyond

Technical brief

Key messages

Monitoring air quality (i.e. collecting continuous and quality data) is integral and critical to detecting air pollutants and to ensuring information sharing with the public, policy-makers and regulatory compliance.

Air quality monitoring is also one of the first and necessary steps for quantifying the health impacts from air pollution and monitoring progress towards reducing air pollution. The main pollutants (e.g. particulate matter [PM], nitrogen dioxide and tropospheric ozone) should be monitored continuously and consistently. Sharing these high-quality and standardized data publicly is essential to raise awareness and encourage decision-makers to take action.

Across and within regions, nations and cities, both air pollution sources and the factors that affect them, such as meteorology and topography, can vary substantially. As a result, the air quality management strategies and associated risks to human health may differ from region to region. Long-term and robust air quality monitoring is required to inform policy decisions and support scientific evidence of the risk of the various atmospheric pollutants. A well-functioning monitoring system will:

- Advance public health work by better characterizing and quantifying exposures where populations reside and across different micro-environments (such as in traffic and industrial hotspots) so that the health sector can understand the health impacts and evaluate the effectiveness of air pollution interventions.
- Enhance understanding of the atmospheric science of air pollution processes across different time periods (e.g. short-term weather vs long-term climate trends) and geographies (e.g. city vs country). These processes can include the chemical composition of source emissions and meteorological factors that influence local and transboundary air pollution.
- Support decision-makers to identify the highest sources of emissions. Inventories of emissions are crucial to understand where the greatest health benefits can be achieved from sectoral interventions.

World Health Organization

Ground-based, reference-grade instruments are the standard method for air quality monitoring, providing robust and continuous observations of air pollutant concentrations. They are, however, costly and require skilled technicians to maintain, so they can be complemented by satellite observations (with a lower spatial and temporal resolution) or low-cost sensors (with lower accuracy). Modelling tools can also complement air quality monitoring, when properly adapted and calibrated for a specific city or region, so as to enable accurate forecasting of air pollution and evaluation of health impacts.

While progress has been made in the past decades with enhanced national monitoring programmes, new technologies and increased data availability, there is still an important fraction of the population, particularly in low- and middle-income countries (LMICs), that is falling behind in tracking their local pollution. These populations where monitoring is inadequate are often the same citizens who are exposed to harmful levels of air pollution. More concerning is that the places without air quality monitoring are often where the most vulnerable populations live and where pollution levels may be highest.

Key concepts

Concentration: The amount of a contaminant present in a certain volume or mass of environmental medium (e.g. air, water) *(1).*

Exposure: The contact between individuals and air pollution, capturing both how polluted the air is and how much time people spend in different environments with different concentrations of pollution.

Monitoring: This involves collecting data continuously, most of the time from measurements but models can support and complement ground-level observations and satellite products.

Scientists conduct air-pollution research in Pretoria, South Africa, analyzing atmospheric conditions and particulate matter levels.
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Overview

Air pollution is recognized as a major environmental risk factor to health – especially for noncommunicable diseases *(2).* World Health Organization (WHO) estimates from 2021 indicate that 4.3 million lives are lost to ambient air pollution from fine PM. Having a good understanding of the magnitude of human exposure to air pollutants and having the legal, policy and scientific tools to monitor progress in protecting health are critical for addressing this risk factor. Air pollution is embedded in the Sustainable Development Goals (SDG) indicators framework through SDG 11.6.2 (Air quality in cities), SDG 7.1.2 (Access to clean household energy) and 3.9.1 (Mortality from air pollution) *(3).*

Monitoring air quality essential for health impact assessment

WHO has been supporting countries to monitor air quality for health impact assessment for decades *(4–8)* with guidance. Monitoring and reporting air quality information is essential to:

- Evaluate compliance with national air quality standards and WHO air quality guidelines (AQG).
- Trigger policy actions (e.g. establishing/updating air quality standards).
- Assess air pollution hotspots, trends and their public health significance (e.g. burden of disease, epidemiological research).
- Guide the actions to mitigate the health effects of air pollution.
- Justify the costs and efforts needed to implement and evaluate the policy (e.g. cost-benefit analyses).
- Mobilize society for policy support.
- Validate the effectiveness of mitigation actions (e.g. accountability studies).

Exposure information is central to assess the health burden of air pollution

Exposure assessment of air pollution is central to evaluate its burden and impacts on public health. The extent of exposure is affected by land use and daily activities, and depends on factors such as the age, occupation and other characteristics of individuals.

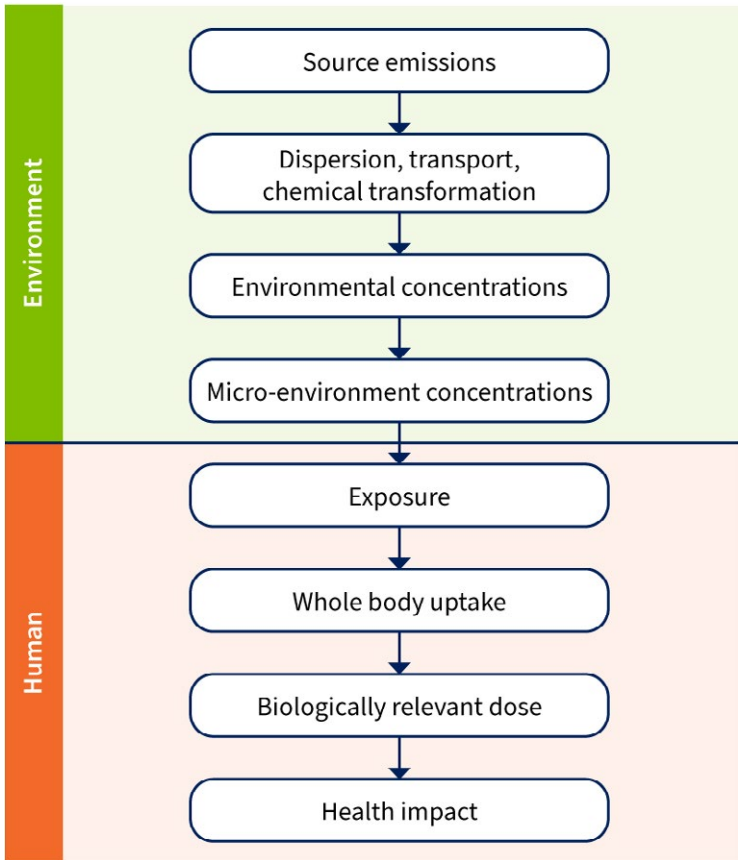
See SPS[1]: *Exposure to health damaging air pollutants*

See SPS: *Land use planning – sectoral solutions for air pollution and health*

[1] Science and Policy Summary.

The data generally used to assess exposure have to be regarded as, and often are, approximations of individuals' exposure to air pollution. Since exposures occur over multiple locations and time periods, it is difficult to accurately estimate "true" exposures at the individual level and for this reason, population-level exposures are commonly used (e.g. post codes or census geographical areas) *(9)*. Ideally, the more measurements of exposures across different micro-environments (e.g. homes, offices, vehicles, cities) in population-level exposure assessments, the greater the likelihood that the estimated population exposure is closer to an individual's "true" exposure. The accuracy and precision of this population exposure may depend on the exposure measurement and/or modelling method used. As shown in Fig. 1, exposure is characterized through the understanding of the sources of air pollutants and also the physical course they take from the source (i.e. the exposure pathway) and the way the pollutant enters the body (i.e. the exposure route).

Fig. 1. From source of air pollutants to health impact



Source: Modified from Nieuwenhuijsen and Brunekreef, 2008 *(10)*.

See SPS: *Health effects of air pollution – evidence and implications*

Global progress in air quality measurement not equal

Since 2011, WHO has compiled air quality ground measurements – mostly from official sources – to inform exposure assessment to estimate the long-term effects of air pollution on health. Since its first release, PM data coverage (i.e. both PM with an aerodynamic diameter less than or equal to 2.5 microns [$PM_{2.5}$] and 10 microns [PM_{10}]) has expanded over sixfold: from 1100 settlements to about 7200 in the 2024 version of the WHO Ambient Air Quality Database and includes data from 123 countries *(11).*

Despite these improvements, the availability of ground measurements of $PM_{2.5}$ in the WHO database is not homogeneous around the globe. Around 68% of measurements come from settlements located in high-income countries and 26% in upper middle-income countries *(12).* However, lower middle-income (6%) and low-income countries (0.06%) contain only 6% of the measurements but represent almost 50% of the global population.

Methods of monitoring air pollution to improve public health

Monitoring is about collecting data continuously, not only via ground stations, but also through satellite products and modelling tools. Reference-grade instruments are commonly used in air quality monitoring networks to collect ground-level air pollution as these instruments are based on standardized scientific procedures and methods allowing the comparison of air pollution from different instruments across the globe. They are the most reliable and accurate instruments for long-term monitoring of air pollution concentrations, which is essential to evaluate compliance with air quality standards, determine trends and assess health impacts from long- and short-term air pollution exposure. These reference-grade instrument networks are valuable because of their precision, and they are an integral part of air quality legislation and governance.

See SPS: *Legislation on air quality and implications for health*

It is also important that local data are collected at diverse population exposures to support the assessment of the effectiveness of interventions. However, countries usually locate most reference-grade instruments that are used for regulatory purposes in urban areas where population density is greatest, while fewer instruments are situated in rural areas or pollution hotspots. Without these local data, it is not possible to objectively evaluate if policies are having any impacts.

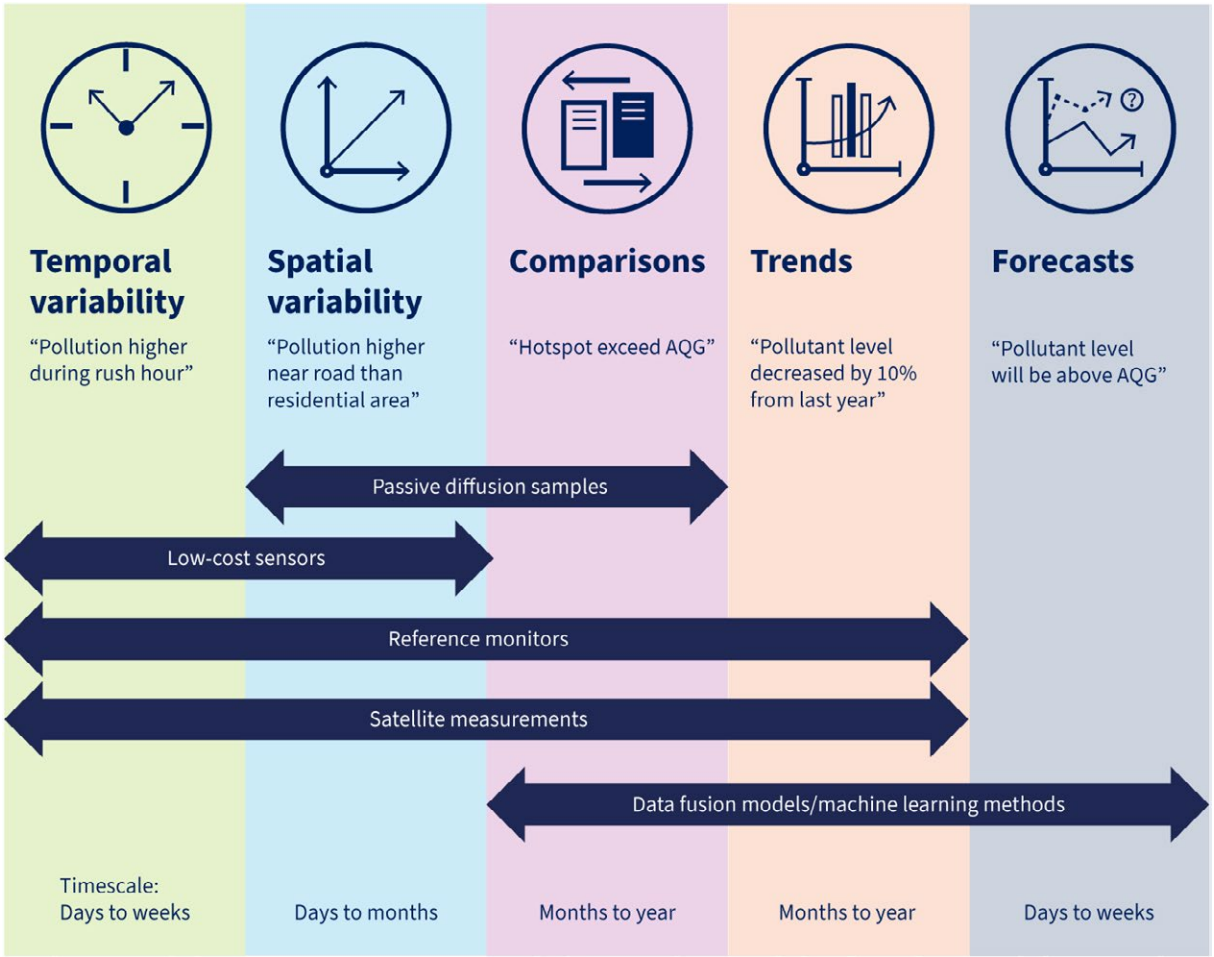
Air quality observational methods from ground level to space

Reference-grade air quality networks are the gold standard for air quality monitoring, using high-quality instruments like tapered element oscillating microbalance (TEOMs) for PM and chemiluminescence for gases, as mandated by regulatory bodies such as the European Committee for Standardization *(13)* and United States Environmental Protection Agency *(14).* While they offer unparalleled accuracy for long-term trend analysis, their high cost and complex operational requirements lead to limited spatial coverage, creating gaps in data, especially at the local level. Their distribution is geographically biased, concentrated primarily in developed countries, though many developing countries are rapidly expanding their networks. Potential gaps in reference-grade monitoring networks, particularly in LMICs *(12),* can be complemented by satellite observations (with a lower spatial and temporal

resolution), air sensors (with lower accuracy, also known as low-cost sensors) and air quality modelling systems – leading to hybrid networks that balance the high data quality of reference monitors with broader spatial coverage (see Fig. 2) *(15).* While air sensors are not suitable for replacing reference-grade instruments, periodic calibration with reference-grade instruments can improve their accuracy *(16).*

Satellites provide long-term, consistent, large-scale coverage that complements ground-based networks and offers significant advantages for air quality monitoring. Satellites enable observation of air pollutants across the globe, including in remote or under-monitored regions, ensuring no area is entirely data scarce. Satellites track air pollution by observing how different gases and particles affect the light and heat coming from Earth's atmosphere. With high temporal and spatial resolution from modern instruments, satellites can capture urban-scale pollution patterns and track the movement of pollutants across borders. Additionally, multiple satellite sensors (e.g. MODIS by NASA, SLSTR and TROPOspheric Monitoring Instrument [TROPOMI] by the European Space Agency) can reliably estimate daily and seasonal surface air pollutant concentration levels across diverse regions when calibrated with ground-level data *(17).* Their long-term, standardized datasets are invaluable for trend analysis, climate-health research and model validation.

Fig. 2. Range of current capabilities and applications across different types of air pollution measurement techniques and modelling tools



Source: WHO; 2023 *(6).*

Air quality modelling methods

While monitoring involves collecting data continuously, models help and complement ground-level observations and satellite products by:

- Interpreting the monitored data (not just emissions, but also atmospheric chemical composition and chemical processes).
- Filling in gaps where no direct measurements exist (for example, data assimilation or fusion techniques, i.e. analysis and re-analysis).

As well as measuring air pollution with different instruments, mathematical models can also be employed to estimate air pollution levels. There are three types of air quality modelling frameworks:

- **Statistical approaches:** These include pattern repetition, autoregressive models and machine learning/artificial intelligence (ML/AI) techniques, trained using historical air quality measurements and potentially other data. While these statistical approaches are not typically used to predict air pollutant concentrations, they allow users to estimate concentrations over a wide range of spatial scales (from 10 m to 10 km) with models being developed for neighbourhoods, cities, continents and globally. Statistical approaches can simulate behaviours that have already occurred in the past, as these patterns are captured in the historical datasets used for the analysis. However, they may become inaccurate or even misleading when applied to new emission sources or unprecedented meteorological conditions, such as those resulting from climate change.
- **Chemical transport models (CTMs):** CTMs are computational tools that can simulate and predict air pollutant concentrations based on emissions of primary pollutants from multiple sectors (e.g. industry, agriculture, household, transport) as well as the physical and chemical processes governing atmospheric dispersion and transformation. They are a key component of current air quality forecasting systems. The main advantages of these modelling systems include their ability to represent fundamental processes at different spatial scales (i.e. local, regional and global domains), test policy scenarios and forecast air quality a few days in advance. However, such models require robust input data (such as meteorological parameters and emissions inventories) and significant expertise and computing infrastructure to use them properly *(18).*
- **Data fusion models:** These integrate information from different sources of measurements and air quality models. These models generally aim to take advantage of the strengths of measurements (good accuracy but limited to the location of the measurement) and models (good spatial and temporal coverage, but biased) by combining information from the different sources. For example, to produce more accurate and spatially complete estimates of $PM_{2.5}$ for SDG 11.6.2, the global model integrates multiple sources including satellite products, ground measurements and CTMs within a statistical framework *(19).* These globally consistent $PM_{2.5}$ estimates are the backbone of estimating the burden of disease from air pollution (SDG 3.9.1) *(20).*

Monitoring should be complemented by emissions inventories and the use of models for an integrated approach to evaluating exposure and health impacts. Emissions inventories aren't models; they are fundamental to understanding source contributions and are essential inputs to modelling. Emissions inventories allow countries to estimate quantities

of primary pollutants that have been or will be discharged into the atmosphere by specific sectors – such as transport, energy production, building, agriculture and waste – enabling the prioritization and design of policies to abate the most serious sources. Inventories should be evaluated by comparing with ambient measurements as the uncertainty of these estimates can be significant, depending on the availability of robust activity data and local emission factors.

A complementary way to assess the comparative importance of various sectors to air pollution, is to use source attribution or source apportionment models – assessing the chemical components of air pollutants captured and tracing them back to sources from different sectors. Broadly, source apportionment approaches are categorized into receptor models and source-oriented (or forward) models. In many cases, especially for secondary pollutants such as ozone, nitrogen dioxide and $PM_{2.5}$, there is no simple linear relationship between emissions and observed concentrations, due to the complex atmospheric chemistry that governs their formation and transformation. In these situations, source-apportionment approaches are crucial to disentangle the contributions of various emission sources and to inform the design of effective and evidence-based emission reduction policies.

Accurate and accessible air quality information is crucial because air pollution is a transboundary challenge that affects health worldwide. Reliable data sharing among countries enables coordinated assessments of pollution exposure and health impacts across regions, fostering transparency and trust. Such collaboration supports harmonized standards, informed policy-making and targeted interventions that transcend national borders.

A satellite monitors air quality from orbit to help protect public health and the environment.
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Barriers to using air quality information for health and beyond

Lack of measurements that capture the exposure of the entire population: National monitoring systems commonly locate reference-grade instruments in densely populated areas. While this is important for assessing regulatory compliance, it is less beneficial for studying the health impacts of the entire population whose exposure can vary significantly from urban to rural areas as well as in different micro-environments (e.g. close to roads or industries, or within households).

Lack of harmonization of data ownership: Accessing national data can be challenging when the responsibility for air pollution monitoring is held at subnational level or distributed between multiple national agencies. In some countries data collection is not centralized and the responsible cities, districts or states may conduct monitoring without national oversight and coordinated reporting which can make data access less effective.

Lack of standardized monitoring and reporting: Another challenge in monitoring air quality is the diversity of methods used to monitor and report air quality, which may be due to lack of or varying operating procedures and technical requirements to determine the acceptability of measurements and insufficient training and sustaining of qualified technical personnel. This is important in regional and global assessments, which require comparable datasets. In addition, data may be lost due to insufficient data coverage and low data quality as a consequence of inappropriate quality assurance or quality control methods.

Enablers to using air quality information for health and beyond

Trust and open access of national data: In addition to measuring air pollutants, it is important that countries share the data with the public in a way that is easy to access and interpret. Publishing air pollutant concentrations supports national institutions in monitoring air quality and conducting health impact assessments; enables researchers to conduct epidemiological studies; and allows the public to understand which pollutants and levels affect them the most. Furthermore, leveraging national data, which have been validated by countries in a clearly defined methodology, to inform regional policy frameworks such as the Convention on Long-Range Transboundary Air Pollution (CLRTAP) can strengthen trust in data and enhance open access to data *(21).*

Multisectoral collaboration: The United Nations Environment Programme (UNEP) latest resolution on air quality saw countries calling for “increased global cooperation to tackle air pollution by sharing best practices, tools, data, and information” *(22).* As a result, in 2024, the UNEP-convened Climate and Clean Air Coalition (CCAC) launched the Air Quality Management Exchange Platform, designed to help air quality professionals everywhere tackle air pollution by providing a one-stop-shop platform providing the latest air quality management guidance and tools from various reference institutions *(23).*

Way forward

UN agencies

Multiple UN agencies and programmes are involved and cooperating on air quality activities, such as monitoring air quality and building institutional capacity to reduce air pollution to protect public health. Close coordination, ensuring synergies and efficiency, avoiding duplication of efforts, and sustainable financing for key UN agencies are needed more than ever to ensure the UN system can support countries in their journey to address air pollution and its impacts.

The World Meteorological Organization is coordinating national meteorological and hydrological services and has been in the forefront of measuring and modelling air pollution through its Global Atmosphere Watch (GAW) research programme *(24)* which focuses on providing integrated products and services related to atmospheric composition of relevance to society. Within this main goal, GAW activities aim to:

- **Maintain and apply long-term global systematic observations** of the chemical composition and related physical characteristics of the atmosphere.
- **Promote best practices on atmospheric composition-related measurements** including international standards for key variables and instruments; and on intercomparisons and calibration campaigns to ensure high-quality standards of the GAW monitoring network.
- **Support the design and implementation of early warning systems** for air pollution hazards.
- **Coordinate, promote and produce scientific assessments** for advancing scientific understanding through analysis of global datasets.
- **Build capacity and promote education** on atmospheric composition information.

UNEP supports governments' environment ministries on air quality monitoring and affordable technologies; provides policy support in key air-polluting sectors; strengthens regional cooperation and knowledge sharing, and the development of robust awareness-raising strategies and advocacy. Key activities include *(25):*

- **Strengthening laws and standards on air quality** by providing advisory services to governments to have measures, policies or frameworks for air pollution control implementation and enforcement.
- **Monitoring air quality actions, policies and programmes** to reduce air pollution across various sectors (such as transport, industry, waste, etc.).

Coordinating the Global Air Quality Cooperation Network strengthen collaborative action, share knowledge, and enhance capacity for air quality management.

Promoting Clean Air Action with various initiatives (International Clean Air Day, Beat Pollution, etc.).

Promoting the CCAC activities such as the Clean Air Flagship *(26)*, and the development of information and guidance to build capacity among air quality managers via the Air Quality Management Exchange Platform, coordinating integrated assessments on air quality and climate actions, financing and funding for clean air and sectoral solutions.

WHO promotes interventions and initiatives for healthy sectoral policies (including energy, transport, housing, urban development and electrification of health care facilities), addressing key risks to health from air pollution indoors and outdoors, and contributing to achieving health co-benefits from climate change mitigation policies. Activities to support technical and policy implementation of measures to prevent the health impacts of air pollution and obtain other health co-benefits include:

Evidence synthesis on health-related topics:

- Normative guidance setting (e.g. WHO air quality guidelines) *(27).*
- Recommendations and development of tools for health risk assessment of air pollution *(5).*

Monitoring and reporting on exposure and health impacts of air pollution through the SDGs:

- Hosting databases on air quality ground measurements and air quality standards as well as on household energy for cooking, lighting and heating.
- Custodial agency for SDG 3.9.1 (mortality from ambient and household air pollution), SDG 7.1.2 (access to clean household energy) and SDG 11.6.2 (air quality in cities) *(28).*

Institutional capacity building and technical support to countries:

- Development of *Air pollution and health training (APHT) toolkit* for health workers, which enables clinical and public health professionals to understand the health risks of air pollution and identify risk reduction measures.
- Development of tools to assess the health, climate and economic impact of air pollution and interventions (e.g. CHEST, AirQ+, CLIMAQ-H, HEAT, GreenUr, IsThAT) *(29).*

Environment sector

The environment sector should focus on legal requirements (such as legislation and air quality standards) that leverage WHO air quality guidelines which are health-based recommendations for key air pollutants. Since reference-grade instruments are well suited to track long-term changes in air pollutants concentrations, they are a key requirement in an air quality monitoring programme. Nations should strive to have at least one reference-grade instrument in an urban centre and one in a rural area in order to assess the national background levels and transboundary influence of air pollution. Countries should also strive to improve the local data collection duration as a longer duration of historical data can allow more robust exposure estimates and health impact assessments. When investing in infrastructure, it is also important to invest in technical human capacity and regional cooperation. It is also critical that these local data be publicly available to citizens to inform their personal behaviours and to academics to support their research on local air quality.

Ensure that the current set of pollutants measured by national air quality monitoring networks includes the chemical composition and size distribution of PM.

Encourage countries to invest in long-term reference-grade strategies so as to enable the tracking of air quality changes and compliance with standards.

Promote open access principles as this can help to speed up advances in health assessments.

Health sector

Surveillance and reporting of health data on mortalities and morbidities should be supported to assess long-term and short-term health impacts of air pollution. Priority should be given to collecting mortality and morbidity data and integrating air pollution-related International Classification of Diseases (ICD) codes to those diseases associated with air pollution exposure. Ideally, an integrated surveillance system with health and environmental exposures should be developed.

It is important to highlight that air quality monitoring networks that represent only urban areas are not suitable for characterizing the exposure of the entire population and may bias the exposure estimates in epidemiological studies *(30).*

Research

Explore research on emerging pollutants, their source contributions and associated health effects.

Explore research and development to enhance instrument performance in diverse geographical settings.

Explore novel statistical methods for assessing exposure and health impacts from air pollution.

Explore multi-hazard assessments to support the development of early warning systems.

Explore research on the identification of climate factors associated with air pollution processes.

Methodology

WHO defined the scope of the document and collaborated with the World Meteorological Organization to prepare the initial draft based on an overview of measurement and modelling methods available for health impact assessments, supplemented by expert advice. Experts in the following fields were consulted: ambient and household air pollution exposure, epidemiology, air pollution policies, and atmospheric measuring and modelling to obtain recommendations for relevant articles, ensuring access to pertinent literature for this document.

The draft underwent peer review by specialists from various research institutes, universities, public organizations and UN agencies. And feedback was addressed by the main contributors. Finally, WHO staff and consultants from the WHO Air Quality, Energy and Health Unit reviewed the report to ensure alignment with the WHO requirements for the collection of WHO Air Quality, Energy and Health Science and Policy Summaries. This series synthesizes current knowledge and evidence on air quality, energy access, climate change links and health, primarily to inform intergovernmental discussions.

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Declarations of interest

All external experts submitted a declaration of interest to WHO disclosing potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the report. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects covered by the report.

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